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RESEARCH ARTICLE



Sociological perspectives on AI ethics in higher education: a comparative study of student experiences in the UAE and Jordan

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ABSTRACT

This cross-sectional study aims to examine the ethical values regulating the use of generative artificial intelligence (ChatGPT) among university students in the United Arab Emirates and Jordan. A descriptive–analytical survey of 1,000 students across multiple faculties revealed a high adoption rate of approximately 82% in both contexts, with no statistically significant differences by university affiliation (odds ratio = 1.009, $p = .955$) or sex. Students primarily reported using ChatGPT for information retrieval, writing assistance, error detection, and summarization. Despite its frequent use, mean scores on ethical-value items indicate strong disagreement with substituting traditional instruction, uncritically trusting AI-generated content, and prioritizing AI over face-to-face interaction. Regression analysis further demonstrates that ethical values significantly predict responsible usage behaviors, suggesting that AI integration occurs within a framework of moral regulation rather than unrestrained technological dependence. Although usage patterns appear structurally similar across the two national contexts, variations in institutional policy environments highlight differing governance dynamics that may shape long-term regulatory approaches. The findings contribute to ongoing debates on AI in higher education by emphasizing not only the continued role of ethical norms in shaping student engagement but also the importance of culturally grounded governance frameworks that balance innovation with responsibility.

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1. Introduction

In recent years, the world has witnessed a profound technological revolution driven by the rapid development of artificial intelligence (AI). These advancements have enabled the creation of systems capable of simulating human intelligence and performing tasks previously considered uniquely human. From medical diagnostics and autonomous vehicles to intelligent tutoring systems and creative digital assistants, AI applications have become integral to modern societies, transforming the ways in which people live, work, and interact (Russell & Norvig, 2021).

AI has generated significant benefits, including improving efficiency in industrial production, supporting decision-making in healthcare, enhancing accessibility for individuals with disabilities, and assisting in military operations and crisis management (Haenlein & Kaplan, 2019). However, it has also raised complex ethical challenges, including concerns related to bias, privacy, accountability, transparency, and the broader societal implications of increasing reliance on machine intelligence (Jobin et al., 2019).

In the context of higher education, AI tools such as ChatGPT have opened new possibilities for learning, teaching, and research. Students and educators are increasingly adopting these systems for academic tasks, ranging from writing assistance and language learning to brainstorming and information retrieval (Zawacki-Richter et al., 2019). Although such technologies can enhance educational outcomes, they also raise important questions about academic integrity, fairness, dependency, and the role of human critical thinking.

International frameworks emphasize that educational AI should align with principles of human rights, transparency, and accountability, thereby promoting trustworthy innovation and context-sensitive governance (OECD, 2019; UNESCO, 2021, 2023). Recent policy developments in the European Union classify certain uses of AI in education as high-risk, requiring stricter oversight and institutional accountability. Meanwhile, regional initiatives, such as the United Arab Emirates (UAE) Ethical AI Toolkit, provide practical guidance for the responsible deployment and use of AI technologies (European Parliament, 2025; Smart Dubai, 2019).

These developments underscore the need for robust ethical frameworks to guide AI use in educational environments. Ethics in this context represents not only a set of principles guiding behavior but also a safeguard to ensure that technological innovation aligns with human values such as justice, accountability, and respect for knowledge (Floridi & Cowls, 2019). The absence of robust ethical and legal frameworks risks leaving the future development of AI unregulated and vulnerable to misuse.

Research conducted in the UAE and Jordan highlights a growing interest in educational AI, coupled with concerns regarding accuracy, integrity, and institutional readiness (Abdaljaleel et al., 2024; Abu Hammour et al., 2024; Swidan et al., 2025). Findings from these studies emphasize the urgent need for context-specific ethical frameworks and targeted capacity-building initiatives for both faculty and students. Against this backdrop, the present study offers a comparative analysis of ethical values, usage motivations, and responsible use behaviors among university students in the UAE and Jordan, providing timely insights to inform evidence-based policy and practice.

The study adopts a descriptive-analytical design to: (a) estimate adoption rates and the frequency of ChatGPT use among students in both the UAE and Jordan; (b) analyze variations across different university contexts; and (c) examine cognitive, social and

psychological, and practical motivations, along with ethical dimensions – such as credibility, integrity, bias awareness, confidentiality, critical thinking, and dependence – associated with responsible AI use in higher education (Floridi & Cows, 2019; UNESCO, 2023). By aligning institutional policy discourse with students' reported behaviors and values, the study aims to inform the development of practical guidelines for disclosure, assessment design, and AI literacy initiatives that promote a balanced integration of innovation and accountability within regional higher education systems (UNESCO, 2023; Zawacki-Richter et al., 2019).

1.1. *The dialectical relationship between education and artificial intelligence (AI)*

The relationship between education and AI is inherently dialectical, characterized by both opportunities and challenges. On the one hand, AI offers promising avenues for enhancing the learning process through personalized instruction, adaptive assessment, and real-time feedback. Tools such as ChatGPT enable students to summarize texts, generate ideas, and practice language skills, while also assisting instructors in managing administrative tasks and evaluating performance more efficiently. As Kasneci et al. (2023) note, AI has the potential to foster innovation, creativity, and inclusivity within higher education.

Recent research indicates that the use of ChatGPT is positively associated with enhanced cognitive skills, including thought organization, vocabulary retention, and analytical abilities, while also increasing motivation and engagement in writing tasks (Crompton & Burke, 2023). On the other hand, this integration raises critical concerns regarding ethics, academic integrity, and the risk of overreliance. Students who depend heavily on AI-generated outputs risk undermining their critical thinking, originality, and self-directed learning (Dwivedi et al., 2023). Emerging research raises concerns about cognitive offloading and reduced deep processing when students rely excessively on generative AI tools (Dwivedi et al., 2023; Rudolph et al., 2023).

The absence of robust ethical frameworks complicates the regulation of AI in education, reinforcing tensions between its perceived benefits and potential harms (Dwivedi et al., 2023).

Resolving this tension requires institutions to establish clear ethical guidelines, pedagogical strategies, and regulatory frameworks to ensure the responsible use of AI. Rather than replacing educators or diminishing student agency, AI should be integrated as a complementary tool that enhances both teaching and learning. Zawacki-Richter et al. (2019) emphasize that the future of AI in education depends largely on universities' ability to adopt holistic strategies that address both pedagogical opportunities and ethical challenges, allowing AI to serve as both a catalyst for educational transformation and a subject of critical reflection.

1.2. *Global and policy frameworks*

International organizations have provided guiding principles for ethical AI use. UNESCO's Recommendation on the Ethics of AI (UNESCO, 2021) emphasizes human rights, transparency, and accountability, while the Organisation for Economic Co-Operation and Development's AI Principles (OECD, 2019) promote trustworthy innovation aligned

with democratic values. The European Union recently introduced the AI Act (European Parliament, 2025), which classifies educational AI systems as ‘high risk,’ requiring strict governance. In the UAE, Smart Dubai issued the Ethical AI Toolkit (Smart Dubai, 2019), offering practical guidelines for responsible AI deployment in institutions. While the UAE National Strategy for Artificial Intelligence 2031 emphasizes building AI-ready talent and governance frameworks, Jordan’s Artificial Intelligence Strategy and Implementation Plan of 2023–2027 focuses on accelerating social and economic development through AI systems nationally, emphasizing education and capacity building (Ministry of Digital Economy and Entrepreneurship, 2023). The Arab League Educational, Cultural and Scientific Organization recently adopted the first AI ethical code to protect Arab values in higher education, establishing clear principles for responsible and equitable AI use (Floridi & Cows, 2019; Jobin et al., 2019). These frameworks collectively establish an evolving regulatory landscape, transforming AI ethics from aspirational guidance to enforceable standards with culturally sensitive implementations.

1.3 Adoption and uses

Empirical studies indicate widespread student adoption of AI tools, with learners primarily employing ChatGPT for brainstorming, summarization, and writing support (Ravšelj et al., 2025; Zawacki-Richter et al., 2019). In the UAE, research validating Technology Acceptance Model Edited (TAME)-ChatGPT factors of perceived usefulness, perceived risks, and social norms demonstrates high uptake, highlighting the necessity of explicit academic integrity policies (Sallam et al., 2024; Swidan et al., 2025). Similarly, in Jordan and broader Arab contexts, enthusiasm for AI coexists with concerns regarding accuracy, bias, and integrity, consistent with Technology Acceptance Model (TAM)-driven adoption patterns observed globally (Abu Hammour et al., 2024; Abdaljawel et al., 2024).

1.4. Challenges of AI detection and assessment

A growing body of research has highlighted the challenges of relying on AI detection tools in academic settings. Liang et al. (2023) demonstrate that many Generative Pre-trained Transformer (GPT)-detection systems exhibit bias, particularly against non-native English writers, raising fairness and equity concerns. Perkins (2024) further reveals that such detectors can be easily bypassed with minor text modifications, undermining their reliability as safeguards for academic integrity. Institutions have been warned against over-reliance on detection software, with recommendations for a shift toward assessment redesign, encouraging approaches such as oral defenses, reflective portfolios, and transparent disclosure of AI use (UNESCO, 2023). False positive rates in AI-detection tools, despite often claimed to be low by companies, can have serious academic consequences for accused students.

Research further reveals that AI detection systems disproportionately flag content produced by non-native English speakers or students with distinctive writing styles, potentially tarnishing reputations based on flawed detection results (Giray, 2024). Recent research highlights significant challenges in enforcing AI disclosure policies, with Gonsalves (2024) finding that 74% of students at King’s Business School failed to declare AI use despite mandatory requirements. Using the Theory of Planned Behavior, the study revealed barriers such as

fear of repercussions, unclear guidelines, inconsistent enforcement, and peer influence, underscoring the gap between policy design and actual student compliance.

To address these challenges, scholars and institutions are increasingly advocating for ethical and pedagogical shifts. Floridi and Cows (2019) provided a conceptual foundation, proposing five universal AI ethics principles – beneficence, non-maleficence, autonomy, justice, and explicability – to guide universities in developing ethical assessment practices that prioritize fairness and accountability over punitive measures. In line with this framework, universities are adopting innovative strategies such as portfolio-based evaluations, process-oriented assessments, and AI disclosure forms to promote transparency and ethical AI use (University of Waterloo, 2024). Collectively, these studies emphasize that ensuring academic integrity in the age of AI requires rethinking assessment design, prioritizing fairness, transparency, and student engagement over exclusive dependence on detection tools.

1.5. Research gaps and contribution of the current study

Despite the growing body of literature on the integration of AI, specifically ChatGPT, in higher education, several important gaps remain. First, most existing studies have been conducted in Western contexts (Crompton & Burke, 2023; Dwivedi et al., 2023; Kasneci et al., 2023), with limited empirical evidence from Arab countries. Although UNESCO (2021), OECD (2019), and the EU AI Act (2025) outline global ethical standards, few studies investigate how these principles translate into culturally specific higher education settings such as the UAE and Jordan. This creates a gap in understanding how local values, traditions, and institutional policies shape students' perceptions of ethical AI use.

Second, although AI adoption studies in the region (e.g. Abu Hammour et al., 2024; Sallam et al., 2024; Swidan et al., 2025) have validated factors related to usefulness, risks, and social norms, they rarely examine ethical values directly as a distinct construct. Instead, ethics is often treated implicitly, as a byproduct of academic integrity concerns or technical detection challenges, rather than being studied systematically.

Third, prior research has focused on policy frameworks or the technological reliability of detection tools (Giray, 2024; Liang et al., 2023; Perkins, 2024), with few studies examining large-scale empirical evidence on how students themselves perceive and embody ethical responsibilities in their everyday use of generative AI. This neglects the lived experiences and practical dilemmas of the main stakeholders: university students.

Therefore, the present study seeks to fill these research gaps in several ways: It is one of the first large-scale cross-sectional investigations ($n = 1,000$) in the Arab region that empirically explores the ethical values governing ChatGPT use in higher education. By focusing on both the UAE and Jordan, it provides a comparative perspective between two countries that share cultural similarities but have different AI strategies and regulatory frameworks. Unlike earlier works that emphasize detection or adoption, this study centers explicitly on the ethical dimension, offering insights into student attitudes, behaviors, and perceived responsibilities.

Finally, by integrating quantitative findings with policy frameworks, this study advances the discourse from descriptive adoption patterns toward a sociologically grounded understanding of ethics in AI use, thereby providing practical implications for universities and policymakers in the Arab world to design culturally sensitive guidelines and pedagogical strategies that balance innovation with responsibility.

1.6. Ethical considerations and context-specific insights

Several global and regional studies have addressed the ethical challenges of AI integration in higher education. UNESCO's (2023) Guidance for Generative AI in Education and Research provides a comprehensive framework for policymakers and universities, emphasizing transparency, capacity-building, and ethical alignment with human values.

Zawacki-Richter et al.'s (2019) systematic review concluded that while AI applications in education are expanding rapidly, ethical dimensions remain underexplored. Holmes et al. (2021) reinforced this gap, calling for stronger ethical safeguards in AI-based educational practices. More recently, Kasneci et al. (2023) examined the educational potential of ChatGPT, emphasizing both opportunities for personalized learning and risks of plagiarism, fairness, and over-reliance on automated systems. In a large-scale global study, Ravšelj et al. (2025) surveyed more than 23,000 students across 109 countries, reporting that they primarily use ChatGPT for brainstorming and summarization but also highlighting concerns regarding academic integrity. These findings collectively underscore the global consensus that ethical frameworks are essential to guide AI adoption in higher education.

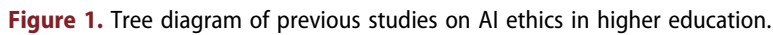
In the Arab region, recent studies have provided more localized insights. Sallam et al. (2024) validated the TAME-ChatGPT model in UAE universities, demonstrating that adoption is shaped by perceived usefulness, ease of use, and perceived risks. Swidan et al. (2025) reported that nearly 80% of students used AI tools, with persistent concerns regarding academic honesty. In Jordan, Abu Hammour et al. (2024) studied Jordanian medical students and found a balance between enthusiasm for ChatGPT and concerns about accuracy, bias, and integrity. Abdaljaleel et al. (2024) examined 2,240 students across Arab countries including Jordan, reporting that attitudes depend on perceived utility and risks, confirming the cultural relevance of TAM-based models, while Al-Hamlan and Al-Said (2023) highlighted the fact that educators across Arab universities lack sufficient training on ethical AI use. Recent investigations reveal that faculty perceptions of academic integrity policies addressing traditional and generative AI plagiarism show significant disparities, as both are rated only as moderately effective (Alsharefeen & Al Sayari, 2025). Faculties demonstrate preferences for educative over punitive approaches while identifying workload constraints, insufficient institutional support, detection challenges, cultural leniency, and systemic limitations as barriers to policy enforcement (Figure 1).

1.7. Linking ethical dimensions to behavior

Ethical dimensions can be mapped onto observable responsible use behaviors, bridging theory and practice. Table 1 summarizes these linkages, integrating ethical principles with concrete actions in higher education.

Table 1 demonstrates that the ethical dimensions of responsible AI use are closely aligned with corresponding observable behaviors in higher education. These mappings connect ethical orientations with measurable practices, consistent with contemporary regulatory expectations (Floridi & Cowls, 2019; UNESCO, 2023; EU AI Act oversight emphasis), forming the foundation for testing hypotheses about responsible AI use.

Recent research further demonstrates that students' perceptions of AI's critical thinking capabilities directly influence their reliance on and interaction with these tools. Studies employing the Perceived Critical Thinking Disposition of Generative Artificial Intelligence (PCTD-GAI) scale reveal moderately positive perceptions across multiple



1.8. Synthesis and research gap

Despite these insights, gaps remain. Most studies address adoption or ethics separately, with limited integration of ethical frameworks into observed behaviors, thereby highlighting the urgent need for context-specific ethical guidelines that balance innovation with accountability in the UAE and Jordan. These studies also emphasize that ensuring academic integrity in the AI era requires rethinking assessment methods focusing on ethical frameworks, transparency, and pedagogical redesigns rather than relying solely on detection technologies. Region-specific evidence on responsible use behaviors, particularly in Middle Eastern educational contexts, is sparse. Furthermore, regulatory

Ethical Dimension	Observable Responsible Use Behaviors
Credibility	Cross-checking AI outputs for accuracy
Honesty/Integrity	Disclosing AI assistance in assignments
Bias Awareness	Reviewing AI outputs for harmful stereotypes or representational imbalance
Confidentiality	Avoiding sharing sensitive or personal data with AI tools
Critical Thinking	Substantive modification or transformative editing instead of direct copying
Dependence	Reduced independent reasoning and non-disclosure of AI use (negative association with responsible use)
Transparency/ Compliance	Honest declaration of AI assistance despite potential consequences

and pedagogical guidance often lacks empirical grounding in student practices, resulting in the limited exploration of the interplay between ethical principles, technology acceptance, and actual responsible use behaviors. To address these gaps, the present study links TAM/TAME constructs with ethical dimensions to examine both adoption patterns and measurable responsible use behaviors, providing evidence to inform policy, pedagogy, and AI governance in higher education. The study's comparison of the UAE with Jordan contributes to understanding how different institutional infrastructures and policy environments influence ethical AI use in higher education.

1.9. Theoretical framing: universities as moral-regulatory institutions

Building on the identified research gaps, this study adopts a sociological framework that situates AI adoption within broader processes of institutional moral regulation. Although TAM/TAME explain behavioral adoption through perceived usefulness, ease of use, and social influence, they offer limited insight into how ethical norms are collectively structured and sustained within educational environments. To address this limitation, this study draws on Durkheim's concept of moral regulation, conceptualizing institutions as sites where shared norms and boundaries shape acceptable conduct.

From a Durkheimian perspective, universities are not only knowledge-producing organizations but also moral-regulatory institutions that define legitimate academic practices through rules, codes of integrity, assessment norms, and shared expectations. The diffusion of generative AI tools such as ChatGPT, therefore, represents both a technological innovation and a transformation in the normative structure of academic life. Patterns of responsible uses such as disclosure, verification, and modification – can be interpreted as expressions of moral integration, while overreliance or uncritical acceptance may reflect weakening normative coordination.

Anchoring the study in this framework allows ethical-value dimensions (credibility, integrity, bias awareness, confidentiality, critical thinking, and dependence) to be examined as not only individual attitudes but also indicators of how institutional contexts mediate AI engagement. Comparing the UAE with Jordan further enables the analysis of how variations in infrastructure, policy clarity, and institutional regulation influence the density of moral norms surrounding AI use.

Guided by this integrated framework – linking technology acceptance constructs with Durkheimian moral regulation – this study proceeds to examine adoption patterns, ethical-value dimensions, and responsible use behaviors among university students in the two countries.

2. Materials and methods

2.1. Research questions

1. What is the prevalence and frequency of ChatGPT use among university students in the UAE and Jordan?
2. How do adoption rates and usage frequency vary by university (UAE vs. Jordan), sex, and age?

3. To what extent do ethical values predict responsible use behaviors of ChatGPT?
4. How does the duration of ChatGPT use differ across demographic groups and institutional contexts?

2.2. Hypotheses

- **H1:** No significant difference exists in adoption rates between students in the UAE and Jordan.
- **H2:** No significant differences exist in adoption rates by sex.
- **H3:** Age has a limited effect on adoption rates but may influence usage frequency and duration.
H3a: Age has a limited effect on adoption.
H3b: Age has a limited-to-weak effect on frequency and duration.
- **H4:** Ethical values significantly predict responsible use behaviors of ChatGPT among university students.

2.3. Research design

A cross-sectional descriptive–analytical survey design was employed to examine patterns of ChatGPT use and the ethical values regulating AI engagement among university students in the UAE and Jordan. This design allows for both descriptive assessment of usage trends and inferential examination of relationships between ethical orientations and responsible AI behaviors.

2.4. Population and sample

The study targeted students enrolled at the University of Sharjah (UAE), Yarmouk University (Jordan), and Princess Sumaya University for Technology (Jordan). A two-stage cluster sampling procedure was implemented. Lecture sections across faculties (humanities, STEM, business, health sciences) were randomly selected, proportional to enrollment. All consenting students within the selected sections were invited to participate. Inclusion criteria included students aged 18 years or above who were currently enrolled. Exclusion criteria were students with audit status and those who failed to provide consent to participate. The final sample included 1,000 students, providing adequate statistical power for multivariate analysis.

2.5. Data collection tool

A structured questionnaire was developed based on validated instruments and prior empirical research on AI use in higher education (e.g. Sallam et al., 2024; Swidan et al., 2025). The instrument included items on:

1. **Demographics:** age, sex, academic level, field of study.
2. **Adoption:** binary measure (Yes/No).

- 3. **Frequency of Use:** ordinal categories (daily, weekly, monthly, several times per year).
- 4. **Duration of Use:** continuous measure (typical session length).
- 5. **Ethical Values Index:** eight Likert-scale items capturing students’ normative orientations toward AI use, including integrity, credibility, confidentiality, bias awareness, critical engagement, and responsible authorship.

All Likert-scale items were coded as 1 = Strongly Disagree and 5 = Strongly Agree. No reverse-coded items were used in computing composite indices. Higher mean scores reflect stronger agreement with the measured construct. Composite scores were calculated by averaging the eight items.

These ethical dimensions, rather than being considered as separate predictors, were conceptually integrated into a unified Ethical Values Index, reflecting the overall moral-regulatory framework guiding AI engagement. Internal consistency analysis demonstrated strong reliability: Cronbach’s $\alpha = .895$ Table 2.

The questionnaire items were not adopted from a single pre-validated scale; rather, they were developed based on established ethical and AI governance frameworks. As shown in Table 3, the Responsible Use composite reflects the integration of ethical orientations, behavioral standards, and institutional controls.

2.6. Data analysis

Composite indices were computed prior to conducting regression analysis.

2.7. Descriptive statistics

Frequencies, percentages, means, and standard deviations were calculated to summarize demographic characteristics and AI usage patterns.

2.8. Inferential statistics

Multiple regression analysis was conducted to examine the predictive role of the Ethical Values Index on responsible AI use behaviors. Regression assumptions (normality, homoscedasticity, multicollinearity) were assessed prior to model estimation. Model fit was evaluated using:

Table 2. Item–total correlations for ethical values index.

Item	Item–Total Correlation (r)
The app is not prejudiced toward subjects	.757**
I rely on honesty and integrity	.664**
I trust that the program protects personal data	.834**
I substitute traditional material	.799**
I prefer the program because of empathy	.803**
Information is accurate and objective	.832**
I use the program only for assistance	.672**
I maintain critical thinking	.722**

All correlations significant at $p < .01$.

Table 3. Measurement of study constructs.

Construct	Questionnaire Section	No. of Items	Scale	Reliability	Theoretical Basis
Standards of AI Use	Part 3	8	5-point Likert	$\alpha = .88$	Adapted from AI responsible use literature and academic integrity frameworks (Floridi & Cowls, 2019; UNESCO, 2023)
Ethical Values Index	Part 4	8	5-point Likert	$\alpha = .89$	Based on AI ethics principles (Floridi & Cowls, 2019; OECD, 2019; UNESCO, 2021)
Institutional Controls	Part 5	7	5-point Likert	$\alpha = .80$	Informed by policy and governance frameworks (EU AI Act, 2025; Ministry of Digital Economy and Entrepreneurship, 2023; Smart Dubai, 2019)
Responsible Use Composite	Parts 3 + 4 + 2 items from Part 5	20	5-point Likert	$\alpha = .75$	Integrative construct combining ethical orientations and behavioral compliance within a moral-regulatory framework (Durkheim, 1893/2014; Floridi & Cowls, 2019)

- R^2
- Adjusted R^2
- F-statistics
- Significance level ($p < .05$)

2.9. Ethical considerations

This study complied with research ethics standards in both the UAE and Jordan. Ethical approval was obtained from the Research Ethics Committee in Sharjah university (Ref No: REC-23-08-15-01-F), and all participants provided informed consent. Confidentiality and anonymity were ensured, and the collected data were used strictly for academic and research purposes.

3. Results

As shown in Table 4, a binary logistic regression analysis indicated that gender did not significantly predict ChatGPT adoption ($B = .145$, $SE = .163$, $p = .374$). The odds ratio ($\text{Exp}(B) = 1.16$) suggests no meaningful difference in adoption likelihood between male and female students. Therefore, H2 is supported. Model fit was evaluated using:

- $\chi^2 (1) = 0.79$
- $p = .374$
- Nagelkerke $R^2 = .002$

Table 4. Logistic regression predicting ChatGPT adoption by sex.

Variable	B	SE	Wald	$\text{Exp}(B)$	p
Sex (Male = 1)	.145	.163	0.79	1.16	.374
Constant	.820	.214	14.67	2.27	< .001

Note: B = logistic regression coefficient; SE = standard error; Wald = Wald chi-square test statistic; $\text{Exp}(B)$ = odds ratio; p = significance level.

Table 5. Primary data: general knowledge of the program and duration of use.

Item	Category	Frequency	Percentage (%)
Sex	Male	420	42
	Female	580	58
Age (years)	< 20	120	12
	20–25	700	70
	> 25	180	18

Table 5 reveals a slight predominance of female participants (58% compared to 42% males). This indicates that women more actively responded to the survey, possibly reflecting greater interest or availability among female students within the studied population. Nevertheless, the sex distribution remains sufficiently balanced to ensure diversity in perspectives and minimize bias.

In terms of age, the largest group of respondents falls within the 20–25 age range (70%), suggesting that most participants are in their early university years. Students younger than 20 years accounted for only 12%, while those older than 25 represented 18, reflecting the typical composition of undergraduate and early postgraduate cohorts, thereby confirming that the survey effectively captured its intended target group of higher education learners.

Table 6 shows clearly no significant difference in the use of AI tools (such as ChatGPT) between university students from the UAE and Jordan. In both institutions, approximately 82% of respondents reported using AI tools, while about 18% did not.

This near-identical adoption rate suggests that university affiliation is not a statistically significant determinant in the likelihood of AI usage. Instead, the findings reflect a broader trend: the growing recognition of AI as a valuable tool in higher education.

Recent studies have highlighted similar patterns across universities worldwide, with AI integration in learning and academic practices becoming increasingly normalized. For example, a study by Dwivedi et al. (2023) emphasized that students across diverse cultural and institutional contexts show comparable enthusiasm for AI adoption, driven by its perceived benefits in productivity, learning support, and access to knowledge.

The parity in adoption rates across universities in both countries reveals that the importance of AI in higher education transcends institutional boundaries, reflecting a global movement toward embracing AI as an essential component of academic life.

Table 6. Economic and social differences between artificial intelligence in the UAE and Jordan.

University	Yes (Used AI)	No (Did not use AI)	Total	Yes Percentage (%)	No Percentage (%)
University of Sharjah	443	97	540	82	18
Jordanian Universities	378	82	460	82	18

Table 7. Adoption of artificial intelligence between the UAE and Jordan.

Variable	Coefficient (B)	Odds Ratio	p-value
Constant	1.519	4.567	0.000
University (Jordan vs. Sharjah)	0.009	1.009	

Table 8. Logistic regression predicting AI adoption by sex.

Variable	<i>B</i>	Odds Ratio	<i>p</i> -value
Constant	1.517	4.56	.000***
Sex (Female vs. Male)	0.042	1.043	.612

Dependent variable = AI adoption (1 = Yes, 0 = No).

Reference category = Male.

The regression analysis presented in Table 7 reveals that university affiliation (Jordan vs. Sharjah) has no statistically significant effect on students' likelihood of using AI tools such as ChatGPT ($OR = 1.009$, $p = .955$). This indicates that adoption rates are nearly identical across both contexts, reflecting the regional normalization of AI use in higher education. This finding reinforces the idea that the global adoption of AI reflects its increasing integration in academic learning.

As Table 8 reveals, gender does not have a statistically significant effect on AI adoption ($OR = 1.043$, $p = .612$). This indicates that male and female students demonstrate comparable likelihoods of using ChatGPT. Therefore, H2 is supported, confirming the absence of significant sex-based differences in AI adoption.

As shown in Table 9, most respondents (54.5%) became familiar with ChatGPT through self-discovery. This finding underscores a central characteristic of the current generation of students: a strong passion for independent exploration and a curiosity-driven approach to learning. The fact that more than half initiated their engagement with AI tools autonomously reflects a generational shift toward self-directed learning and digital literacy.

Simultaneously, social media accounted for 20.5% of responses, making it the second most influential channel for spreading awareness about ChatGPT. This highlights the powerful role of social media platforms in shaping educational practices, especially in promoting the diffusion of technological innovations among youth. Studies by Selwyn (2022) and Dwivedi et al. (2023) have emphasized how social media acts as a catalyst for educational technology adoption, serving as both an information hub and a motivational driver.

Peer recommendations (12.7%) and university channels (11.7%) also contributed to awareness, although their impact was noticeably smaller compared to that of self-knowledge and social media. This suggests that traditional academic institutions and peer networks are no longer the dominant forces in knowledge acquisition; instead, today's students rely on digital ecosystems and personal curiosity as primary sources.

Overall, these results reinforce the current generation's eagerness to learn, experiment, and innovate on their own, while simultaneously being deeply influenced by the ubiquitous presence of social media. This dual pathway reflects both self-directed learning and digital influence. It also illustrates the democratization of knowledge, but also points to

Table 9. Knowledge of the program and its handling.

Source	Frequency	Percentage (%)
Self-Knowledge	544	54.45
Social Media	212	20.52
Friend's Recommendation	127	12.71
College/University	117	11.71

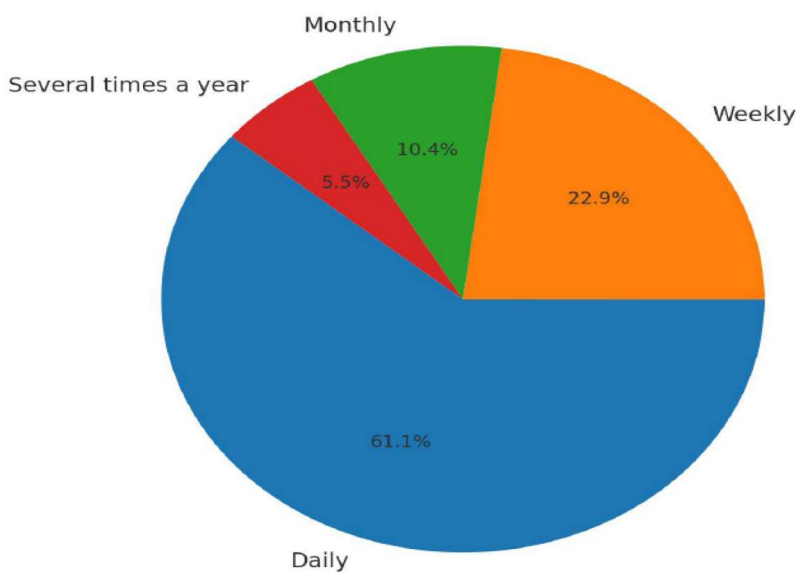


Figure 2. Usage rate of ChatGPT.

the increasing need for higher education institutions to adapt to students’ evolving learning behaviors.

The findings confirm that the entry point into AI usage is largely self-initiated, reflecting students’ enthusiasm and motivation to explore new tools. Simultaneously, the 20% influence of social media highlights its undeniable role as a modern educational medium, aligning with recent studies that stress the integration of digital platforms into higher education strategies (Figure 2).

As depicted in Table 10, age had no significant effect, as age groups did not explain variations in duration of AI use. However, university had an almost significant effect ($p \approx 0.051$). Students at the University of Sharjah tended to report slightly longer usage durations than did students in Jordan, although the effect was weak. The overall model explains 0.5% only of the variance; that is, other personal and contextual factors (motivation, major, access to AI resources, digital readiness) were revealed to play a much larger role.

This aligns with studies by Zawacki-Richter et al. (2019), which highlighted that AI adoption in higher education is mainly driven by self-directed learning and institutional support, rather than by demographic factors alone.

The data in Table 11 reveal that 58.6% of students use ChatGPT daily, followed by 22% weekly, 10% monthly, and 5.3% several times a year. This pattern suggests that ChatGPT

Table 10. Duration of use of the program depending on age and university.

Variable	B	Std. Error	Beta	t	Sig. (p)
(Constant)	2.145	0.210	—	10.21	0.000***
Age_code	0.032	0.078	0.018	0.41	0.679 (ns)
Uni_code	0.187	0.096	0.061	1.95	0.051 †

• *** $p < .001$, † $p \approx .05$, ns = not significant.
• Dependent variable = Duration of ChatGPT use.

Table 11. Program reliability.

Usage Rate	Frequency	Percentage (%)
Daily	612	58.6
Weekly	230	22.0
Monthly	105	10.0
Several times a year	53	5.3

Table 12. Reported purposes for using ChatGPT in education.

No.	Item	Mean	SD	Rank
1	Provide information and respond to inquiries.	3.10	1.00	1
2	Identify grammatical and spelling mistakes.	2.98	1.03	2
3	Text summarization and rephrasing.	2.82	1.08	3
4	Identify mistakes and security flaws in the code.	2.80	1.03	4
5	Help with article writing.	2.75	1.07	5
6	Improving linguistic abilities through discussion.	2.70	1.10	6
7	Generate thoughts and key points for a given topic.	2.68	1.13	7
8	Support and help are available 24 h a day, 7 days a week.	2.67	1.07	8
9	enable pupils determine answers	2.60	1.13	9
10	Material and course registration queries.	2.52	1.14	10

has become deeply embedded in students' academic routines, not merely as an occasional tool, but as a regular component of study practices. Its frequent use aligns with emerging empirical evidence indicating rapid normalization of generative AI in higher education contexts (Dwivedi et al., 2023; Kasneci et al., 2023; Ravšelj et al., 2025).

The high usage frequency highlights a significant shift in how students approach their studies – embracing AI not as optional, but as essential. This supports the notion that Gen Z students are not passive consumers, but strategic users who leverage AI as a productivity enhancer. Their reliance on ChatGPT reflects both the intensity of their academic challenges and their comfort with digital tools – a dynamic that higher education systems must acknowledge and adapt to pedagogical designs in the future.

As presented in Table 12, the mean scores range from 2.52 to 3.10 on a five-point Likert scale. These values indicate moderate levels of agreement regarding students' reported purposes for using ChatGPT in educational contexts.

The item 'Provide information and respond to inquiries' obtained the highest mean ($M = 3.10$, $SD = 1.00$), reflecting a moderate tendency to use ChatGPT as a source of academic information rather than a strongly endorsed or dominant purpose. Similarly, 'Identify grammatical and spelling mistakes' ($M = 2.98$, $SD = 1.03$) and 'Text summarization and rephrasing' ($M = 2.82$, $SD = 1.08$) also indicate moderate engagement, particularly for writing-related support.

More technical applications, such as identifying coding errors ($M = 2.80$, $SD = 1.03$) and article-writing assistance ($M = 2.75$, $SD = 1.07$), reflect comparable moderate endorsement. Lower-ranked purposes, including administrative uses such as course registration queries ($M = 2.52$, $SD = 1.14$) suggest relatively limited use of ChatGPT for non-academic or peripheral tasks.

Overall, students' use of ChatGPT is moderate and task-oriented, primarily centered on informational and writing support functions. The results do not suggest strong or dominant reliance for any single purpose, but rather a balanced and functional engagement aligned with academic assistance needs.

As presented in Table 13, responses indicate a predominantly favorable orientation toward the integration of ChatGPT within educational settings. Most respondents (68.7% agree; 16.4% strongly agree) reported that its use is consistent with the ethics of teaching and learning. However, 13.2% expressed disagreement, while a small proportion (1.8%) took a neutral stance.

When asked whether ChatGPT should be formally integrated into the educational process, 87.4% responded affirmatively, compared to 12.6% who opposed its integration. These results indicate substantial support for institutional adoption of generative AI tools within academic contexts.

However, responses concerning instructor engagement reveal variation in reported pedagogical approaches. Although 66% of participants indicated that instructors do not provide structured guidance on AI use and primarily focus on task completion, 34% reported that instructors regulate and discuss appropriate ChatGPT usage in coursework. This reflects differences in how AI-related practices are addressed within classrooms.

Collectively, these findings reveal a combination of strong student support for AI integration and uneven perceptions of instructional regulation. The results point to the coexistence of high acceptance of AI tools with variability in reported pedagogical structuring, highlighting the relevance of institutional clarity and instructional alignment in shaping AI use within higher education.

As presented in Table 14, the distribution of responses indicates differentiated patterns of engagement with AI-generated content. Although a majority of respondents

Table 13. Attitudes toward integrating ChatGPT into the educational process.

Dimension / Item	Response Category	Frequency	Percentage (%)
(A) ChatGPT Use Is Consistent with the Ethics of Teaching and Learning	Strongly Disagree	21	2.1
	Disagree	111	11.1
	Neutral	18	1.8
	Agree	686	68.7
	Strongly Agree	164	16.4
	Total	1000	100
(B) Should ChatGPT be Integrated into the Educational Process?	Yes	873	87.4
	No	127	12.6
	Total	1000	100
(C) Instructor Engagement with AI Use	The instructor does not provide structured guidance on AI use and focuses primarily on task completion	659	66
	The instructor regulates and discusses appropriate ChatGPT usage in coursework	341	34
	Total	1000	100

Table 14. AI use practices and ethical regulations.

(A) Use of ChatGPT in Coursework		
Response Option	Frequency	Percentage (%)
Submit the AI-generated response without modification	564	56.5
Modify the AI response before submission	408	40.8
Use the program without evaluation in most courses	26	2.6
Total	1000	100
(B) Need for Institutional Ethical Controls		
Response	Frequency	Percentage (%)
Yes	928	92.9
No	72	7.1
Total	1000*	100*

(56.5%) reported submitting AI-generated responses without modification, a considerable proportion (40.8%) indicated that they modify the content prior to submission. A very small proportion (2.6%) reported using the program without evaluation across most courses.

These findings reflect heterogeneous usage practices. Although a notable segment of students rely on AI outputs as generated, a substantial proportion report revising or adapting the content before submission. The data therefore suggest variations in levels of active involvement in the production process, without presuming the depth or quality of such engagement.

With regard to institutional regulation, the overwhelming majority (92.9%) expressed support for implementing formal ethical controls governing ChatGPT use, compared to 7.1% who opposed such measures. This indicates broad acknowledgment of the need for structured guidance at the institutional level, irrespective of individual usage patterns.

Collectively, these results reveal that ChatGPT is actively integrated into coursework practices, while simultaneously accompanied by high levels of expressed support for regulatory frameworks. This coexistence of practical adoption and endorsement of institutional oversight suggests that AI use is not positioned in opposition to regulation but rather as a practice students perceive as requiring formal structuring within academic systems.

Table 15 reveals mean values ranging between 1.40 and 1.81, indicating overall strong disagreement with the listed items. The lowest mean was recorded for 'Ease of accessing information' ($M = 1.40$, $SD = 0.72$), followed by 'Logical/practical solutions' ($M = 1.45$, $SD = 0.80$) and 'Detailed explanation' ($M = 1.46$, $SD = 0.81$). Relatively higher (although still low) means were observed for 'Empathy toward users' ($M = 1.81$, $SD = 1.25$) and 'Attribution of knowledge to self' ($M = 1.76$, $SD = 1.16$). Overall, the results indicate general disagreement across all proposed standards, with modest variation in responses as reflected in the standard deviations.

Table 16 reveals mean values ranging between 1.40 and 1.81, indicating overall strong disagreement with the listed statements. The lowest mean was observed for 'Substitute for traditional class material' ($M = 1.40$, $SD = 0.72$), followed by 'No prejudice in topics' ($M = 1.45$, $SD = 0.80$) and 'Data is accurate and objective' ($M = 1.46$, $SD = 0.81$). Relatively higher means – yet still within the disagreement range – were recorded for 'Prefer program over face-to-face (empathy)' ($M = 1.81$, $SD = 1.25$) and 'Rely on self-values (honesty, integrity)' ($M = 1.76$, $SD = 1.16$). Overall, the distribution of responses indicates consistent disagreement across all ethical-value items, with moderate variation in responses as reflected in the standard deviations.

Table 15. Standards for using the ChatGPT program.

Item	Mean	Std. Deviation
Brief responses	1.59	0.97
Detailed explanation	1.46	0.81
Logical/practical solutions	1.45	0.80
Ease of accessing information	1.40	0.72
Lack of restrictions	1.56	0.94
Awareness of preferences	1.52	0.89
Few errors (dependable)	1.57	0.94
Attribution of knowledge to self	1.76	1.16
Empathy toward users	1.81	1.25

Table 16. Ethical values guiding students' use of the ChatGPT app.

Item	Mean	Std. Deviation
Trust data fully without adding info	1.57	0.94
Data is accurate and objective	1.46	0.81
No prejudice in topics	1.45	0.80
Substitute for traditional class material	1.40	0.72
Trust program to protect personal info	1.56	0.94
Critical thinking despite reliance	1.52	0.89
Use for assistance; main ideas are mine	1.57	0.94
Rely on self-values (honesty, integrity)	1.76	1.16
Prefer program over face-to-face (empathy)	1.81	1.25

Table 17. Multiple regression predicting responsible use behaviors.

Predictor	<i>B</i>	Std. Error	Beta	<i>t</i>	<i>p</i> -value
Constant	1.214	0.148	—	8.21	.000***
Ethical Values Index	0.462	0.041	0.531	11.27	.000***
<i>R</i>²	Adjusted <i>R</i>²	<i>F</i>	df	<i>p</i>-value	
.282	.281	127.02	1, 998	.000***	

In conclusion, the findings indicate that integrity and reliability continue to function as meaningful normative anchors in students' engagement with AI. Despite the high prevalence of daily use, ethical boundaries appear to remain intact, suggesting that AI adoption occurs within a framework of moral regulation rather than uncritical technological dependence. This pattern aligns with Durkheim's conceptualization of moral norms as structuring forces that sustain social order within institutional contexts.

As presented in Table 17, the Ethical Values Index significantly predicts responsible use behaviors of ChatGPT among university students ($\beta = .531$, $p < .001$). The model explains approximately 28% of the variance in responsible use behaviors ($R^2 = .282$), revealing a moderate explanatory effect. The positive coefficient suggests that higher levels of ethical-value orientation are associated with greater engagement in responsible AI practices, including disclosure, modification, and verification behaviors. These findings provide empirical support for H4.

4. Discussion: moral regulation and responsible AI integration in higher education

Adopting a Durkheimian perspective situates the adoption of generative AI not simply as a technological transformation, but as a shift in the moral regulation of academic life. Durkheim conceptualized moral currents as collective normative forces that structure individual conduct through shared values, institutional rules, and patterns of social integration (Durkheim, 1893/2014, 1897/1951). Universities, in this sense, function as moral-regulatory institutions that define legitimate academic practices and sustain collective standards of integrity.

Within this framework, the widespread and routinized use of ChatGPT observed among students can be interpreted as the emergence of a new digital moral current within higher education. When institutional frameworks – such as academic integrity codes, clear policy guidelines, and structured pedagogical guidance – are clearly

articulated, the integration of AI tools becomes embedded within a regulated moral order. Under such conditions, technological adoption may reinforce rather than undermine collective academic norms.

Conversely, when technological engagement expands more rapidly than regulatory and pedagogical structures, risks of normative instability may arise. Patterns identified in the findings – including ethical ambivalence, uneven institutional guidance, and primarily pragmatic engagement with AI tools – echo what Durkheim described as anomie: a weakening of normative boundaries that guide individual behavior. The challenge for universities, therefore, is not the presence of AI itself, but the capacity of institutions to maintain alignment between technological innovation and moral governance.

The comparative findings between the UAE and Jordan further support this interpretation. Differences observed in patterns of engagement appear to reflect variations in institutional environments rather than inherent differences in student values. Factors such as digital infrastructure, policy clarity, and faculty guidance shape the visibility and strength of collective norms surrounding AI use. In this sense, AI governance should be understood as a process of moral integration, whereby universities reinforce shared academic standards within technologically mediated learning environments.

The empirical results also indicate that ethical orientations significantly predict responsible AI use behaviors. This suggests that students' engagement with generative AI is not driven solely by convenience or technological efficiency but is moderated by normative beliefs about academic responsibility and integrity. These findings challenge deterministic narratives that portray AI adoption as inherently disruptive to educational values. Instead, they reveal a pattern of selective and morally bounded integration within higher education contexts.

From a policy perspective, the findings suggest that responsible AI integration requires a balanced approach combining institutional regulation and ethical awareness. Universities may strengthen responsible AI practices by developing clear guidelines for AI use, integrating digital ethics into curricula, and encouraging transparency in the use of generative AI tools. Such strategies support the development of an academic environment where technological innovation remains aligned with shared educational values.

4.1. Policy and educational implications

Transparent AI-Use Disclosure Practices

Based on the observed patterns of AI use and ethical orientations, universities may consider introducing transparent AI-use disclosure practices in academic assignments. Encouraging students to clearly indicate how generative AI tools were used can promote academic honesty while supporting responsible engagement with emerging technologies.

Integration of AI Engagement into Assessment Design

Because a substantial proportion of students indicated limited modification of AI-generated responses, assessment rubrics could incorporate criteria that reward evidence of critical engagement. For example, requiring students to annotate AI-generated content or provide reflective commentary on revisions could help reinforce active learning and reduce passive dependence on automated outputs.

Bias and Source-Critique Training

Given the low mean score for bias awareness ($M = 1.45$), structured instructional activities focusing on evaluating AI-generated content may be beneficial. Classroom exercises that involve comparing ChatGPT outputs with peer-reviewed sources can enhance students' ability to identify inaccuracies, omissions, or implicit biases, thereby strengthening their critical literacy skills.

Critical Thinking Reinforcement

With relatively low mean scores for independent critical engagement ($M = 1.52$), embedding structured AI-critique activities within coursework may support deeper analytical reasoning. Periodic assignments requiring students to evaluate, challenge, or improve AI-generated responses could foster reflective and autonomous learning practices.

AI Ethics and Integrity Modules

Given the predictive relationship between ethical values and responsible AI use behaviors identified in the regression analysis, integrating short AI ethics modules within university curricula may reinforce constructive engagement patterns.

Context-Sensitive Governance Approaches

Although adoption rates were statistically similar across the UAE and Jordan, institutional policy environments may shape how AI governance evolves over time. Universities should therefore develop context-sensitive regulatory frameworks that balance innovation with accountability.

4.2. Limitations and future research

Despite its contributions, this study has several limitations that should be acknowledged. First, the research relies on self-reported survey data, which may be influenced by social desirability bias or differences in respondents' interpretations of AI use practices. Future studies could complement survey methods with qualitative approaches such as interviews or classroom observations to capture deeper insights into students' actual AI use behaviors.

Second, the study focuses on two higher education contexts – the UAE and Jordan – which may limit the generalizability of the findings to other cultural or institutional environments. Future research could extend the comparative scope by including universities from additional regions or educational systems to explore broader patterns of AI adoption and governance.

Third, the study examines ethical orientations and adoption patterns at a particular moment in time. Given the rapid evolution of generative AI technologies and institutional policies, longitudinal research would be valuable to examine how ethical norms, regulatory frameworks, and patterns of AI use evolve over time.

Overall, future research should continue to investigate the dynamic relationship between technological innovation, institutional governance, and moral regulation within higher education. Such work will be essential for understanding how universities can sustain academic integrity while adapting to the transformative potential of generative AI.

5. Conclusion

This study examined patterns of ChatGPT adoption and ethical orientations among university students in the UAE and Jordan. The findings demonstrate that while generative

AI has become widely integrated into academic routines, its use remains moderated by ethical considerations and institutional norms. Interpreted through a Durkheimian perspective, AI adoption reflects a broader transformation in the moral regulation of academic life rather than a simple technological shift. Strengthening institutional guidance, ethical awareness, and critical engagement will be essential for ensuring that generative AI contribute positively to higher education while maintaining academic integrity.

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Data availability statement

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

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